# Machine Learning Engineer Assessment

Fine Tuning LLaMA-2 for a Domain Specific HR Policy Assistant

Name: Chamindu Nipun

Date: 27/02/2026

# 1 Introduction

This assessment proposes designing and implementing a fine-tuned Large Language Model (LLM) for a domain specific enterprise use case. After reviewing repository on GitHub and analyzing candidate models, I selected LLaMA-2 for fine tuning due to its flexibility, control, and suitability for enterprise-grade customization.

The use case focuses on developing a Domain Specific HR Policy Assistant for Sri Lankan Companies, aimed at accurately answering HR related questions based on local labor regulations and corporate HR policies.

While prompt engineering can guide a general purpose LLM, it has limitations

- Inconsistent responses**:** A model without domain adaptation may provide conflicting or partially correct answers.
- Hallucinations: General LLMs can produce plausible but incorrect information when prompted on specialized HR policies.
- Limited domain knowledge**:** Prompting alone cannot teach the model specific local regulations or company policies.

Fine tuning addresses these challenges by

- Embedding domain specific knowledge directly into the model
- Ensuring consistent, policy aligned responses
- Reducing hallucinations and improving factual accuracy

By combining LLaMA-2's robust architecture with targeted fine-tuning on Sri Lankan HR policies, this approach ensures reliable, accurate, and context-aware assistance for enterprise HR operations.

## 2 Model Selection

Selected Model: LLaMA 2 (LLaMA-2 is an auto-regressive, decoder-only transformer model)

| Criterion | LLaMA-2 | GPT-3.5 Turbo | Business Value Perspective |
|---|---|---|---|
| **Cost Efficiency** | No per-token API cost, deployable on local or cloud GPU | Pay per token API usage | LLaMA-2 reduces operational costs, especially for high volume HR queries, GPT-3.5 Turbo can become expensive over time. |
| **Privacy** | Can host locally, sensitive HR data stays secure and compliant | Data sent to OpenAI servers | LLaMA-2 enables data privacy compliance, protecting sensitive employee information |
| **Fine Tuning Flexibility** | Supports LoRA, PEFT, QLoRA; full dataset customization | Cannot fine-tune directly | LLaMA-2 allows domain-specific adaptation, ensuring HR policies are accurately reflected |
| **Technical Control** | Full access to weights, supports parameter efficient fine tuning (LoRA/QLoRA), custom deployment | API access only, no weight control, only prompt engineering | LLaMA-2 allows enterprise teams to fully customize the model for HR policies, ensuring consistent answers without external dependency |

While GPT-3.5 Turbo offers high performance and easy API based integration, its limitations in model access, architectural customization, and data privacy make it less suitable for enterprise HR applications. In contrast, LLaMA-2 provides full fine-tuning control, secure on-premise deployment options, and flexibility to incorporate domain specific HR knowledge, ensuring consistent, accurate, and policy aligned responses.

# 3 Use Case Definition

**Domain Specific HR Policy Assistant**

## 3.1 Problem Statement

Employees frequently have recurring HR related questions, such as

- What is the maternity leave policy?
- How many casual leaves are allowed?
- Can unused annual leave be carried forward?
- What is the notice period for resignation?

While generic LLMs can provide broad answers, they often fall short in

- Alignment with Sri Lankan labor laws
- Adherence to company specific HR policies
- Maintaining a structured, professional corporate tone

## 3.2 Proposed Solution

To address these gaps, LLaMA-2 will be fine-tuned to

- Accurately answer domain specific HR queries
- Align responses with Sri Lankan labor regulations
- Ensure policy consistent and reliable answers
- Maintain a formal, professional corporate tone suitable for HR communications

## 3.3 Business Value

Implementing a fine-tuned HR Policy Assistant offers significant value to the company

1. Time Savings: Automates responses to frequently asked questions, reducing repetitive workload for HR staff.
2. Consistent HR Guidance: Ensures all employees receive uniform, policy aligned answers, avoiding conflicting interpretations.
3. Risk Reduction: Minimizes the likelihood of incorrect advice regarding labor law compliance or company policies, protecting the company from potential legal or operational issues.
4. Scalability: Supports rapid growth by handling large volumes of HR queries without additional human resource overhead.

By combining LLaMA-2 robust architecture with targeted fine-tuning on Sri Lankan HR policies, the company can provide employees with accurate, consistent, and compliant HR guidance, while improving operational efficiency and reducing compliance risks.

# 4 Data Collection & Preparation

For fine-tuning LLaMA-2 to act as a Domain Specific HR Policy Assistant, preparing a clean and structured dataset is critical. This section explains where the data comes from, how it is organized for fine-tuning, and the steps to clean and preprocess it.

## 4.1 Data Sources

Data for fine-tuning is collected from multiple trusted sources to ensure accuracy and relevance

1. Sri Lankan Labor Law Documents
   - Government published laws and regulations like the Shop and Office Employees Act, Industrial Disputes Act, and Wages Board Ordinances.
   - Provides legally accurate answers for HR related questions.

2. Company HR Policies & Manuals
   - Internal employee handbooks, leave policies, payroll guides, and SOPs from Sri Lankan companies.
   - Ensures responses are aligned with company specific practices.

3. HR FAQs and Historical Queries
   - Collected from internal HR ticketing systems, employee emails, or chat support.
   - Captures real-world questions employees frequently ask.

## 4.2 Converting Raw Data into a Fine-Tuning Dataset

To fine-tune LLaMA-2 for a Domain-Specific HR Policy Assistant, all collected raw HR data must be converted into structured Q&A pairs. This ensures the model learns accurate, policy-aligned, and context aware responses.

**Step 1**

Question Answer Pair Extraction - Identify HR-related questions and corresponding answers from the collected raw data (laws, company policies, FAQs). We can use below methods,

- Manual Extraction
- Automated Extraction Using NLP
- Hybrid Approach

**Step 2**

Structuring in JSON Format - Once Q&A pairs are extracted, they are organized for LLaMA-2 fine-tuning

- Chat Format (for instruction-based LLMs)

```
{
  "messages": [
    {"role": "system", "content": "You are an HR policy assistant for Sri Lankan companies."},
    {"role": "user", "content": "How long is the maternity leave?"},
    {"role": "assistant", "content": "Employees are entitled to 84 days of maternity leave, following company policy and Sri Lankan labor law."}
  ]
}
```

**Step 3**

Quality Checks - Ensure answers match official HR policies and Sri Lankan labor laws. Maintain formal HR tone suitable for corporate communications.

## 4.3 Cleaning & Preprocessing

Before fine-tuning, the dataset is cleaned and preprocessed to improve model performance and reliability

1. De-duplication - Remove repeated or identical Q&A pairs to avoid bias.
2. Normalization - Standardize text formats, numeric values, dates, and leave types.
3. Anonymization - Remove personal employee names or identifiers to ensure privacy and compliance.
4. Consistency Check - Verify all answers against the latest labor laws and HR policies to prevent errors.
5. Text Cleaning - Remove irrelevant text, formatting issues, and special characters from source documents.
6. Segmentation - Split long policy paragraphs into smaller, clear Q&A pairs suitable for model input.

## 4.4 Dataset Size & Split

For efficient LoRA/QLoRA fine-tuning, the dataset will consist of a sufficiently large collection of high quality HR related Q&A pairs to ensure strong domain coverage. The data will be divided into training, validation, and test sets using an 80%, 10%, and 10% split respectively. This approach enables effective learning while ensuring proper evaluation and performance validation.

# 5 Fine-Tuning Methodology

This section explains the technical implementation strategy used to fine-tune Meta AI LLaMA-2-7B-Chat for the Domain-Specific HR Policy Assistant using QLoRA. The methodology is designed to be memory efficient, production-ready, and aligned with enterprise GPU constraints.

## 5.1 Approach Overview

Instead of full fine-tuning (updating all 7B parameters), I apply QLoRA (Quantized Low-Rank Adaptation)

- Base model loaded in 4-bit quantized format (NF4)
- Original weights frozen
- Train only small LoRA adapter layers
- Merge adapters with base model after training

This reduces GPU memory usage, training cost, risk of overfitting while maintaining high domain adaptation performance.

## 5.2 QLoRA Hyperparameters

The following table presents the complete set of hyperparameters selected for this fine-tuning run, with justifications grounded in best practices from the QLoRA research literature and the gpt-llm-trainer repository.

| **Hyperparameter** | **Value** | **Justification** |
|---|---|---|
| **Model Base** | LLaMA-2-7B-chat | Optimized for instruction following, 7B balances performance and GPU memory |
| **Quantization** | 4-bit (NF4) | Reduces GPU memory, enables fine-tuning on 16–24 GB VRAM |
| **LoRA Rank (r)** | 16 | Balances expressiveness and parameter efficiency for domain adaptation |
| **LoRA Alpha (α)** | 32 | standard scaling convention for stable training |
| **LoRA Dropout** | 0.1 | Prevents overfitting on a domain specific HR dataset |
| **Target Modules** | q_proj, v_proj | Query and value projections yield best quality gains per trainable parameter |
| **Learning Rate** | 2e-4 | Standard for QLoRA, pairs with cosine scheduler and warmup |
| **LR Scheduler** | Cosine + Warmup | 10% warmup steps, then cosine decay reduces loss spikes |

| | | |
|---|---|---|
| **Batch Size** | 4 (per device) | Adjusted to GPU VRAM, gradient accumulation × 4 = effective batch 16 |
| **Gradient Accum.** | 4 steps | Simulates larger batch without additional VRAM |
| **Epochs** | 3–5 | Sufficient for domain convergence, monitor val loss to avoid overfit |
| **Max Seq. Length** | 512 tokens | Covers most HR Q&A pairs, longer context increases memory cost |
| **Optimizer** | paged_adamw_8bit | Memory efficient, ideal for QLoRA |
| **Weight Decay** | 0.01 | Regularization to prevent overfitting on small HR dataset |
| **FP16 / BF16** | BF16 | Better numerical stability for Ampere+ GPUs |

## 5.3 Fine-Tuning Workflow

The end to end fine-tuning pipeline follows six sequential phases. Each phase maps directly to code components in the gpt-llm-trainer repository and standard Hugging Face PEFT/TRL libraries.

| Phase | Step | Details |
|---|---|---|
| **1. Setup** | **Environment** | Install transformers, peft, bitsandbytes, trl, datasets, configure GPU (CUDA) |
| **2. Load** | **Base Model** | Load LLaMA-2-7B-chat with 4-bit NF4 quantization via BitsAndBytesConfig |
| **3. Adapt** | **LoRA Config** | Apply LoRA adapters to q_proj and v_proj layers using get_peft_model() |
| **4. Tokenize** | **Dataset** | Format HR Q&A as chat template |
| **5. Train** | **SFTTrainer** | Use TRL's SFTTrainer with TrainingArguments, monitor val loss per epoch |
| **6. Merge** | **Adapters** | Merge LoRA weights into base model and save full model for deployment |

## 5.4 Why This Method is Suitable for CDAZZDEV

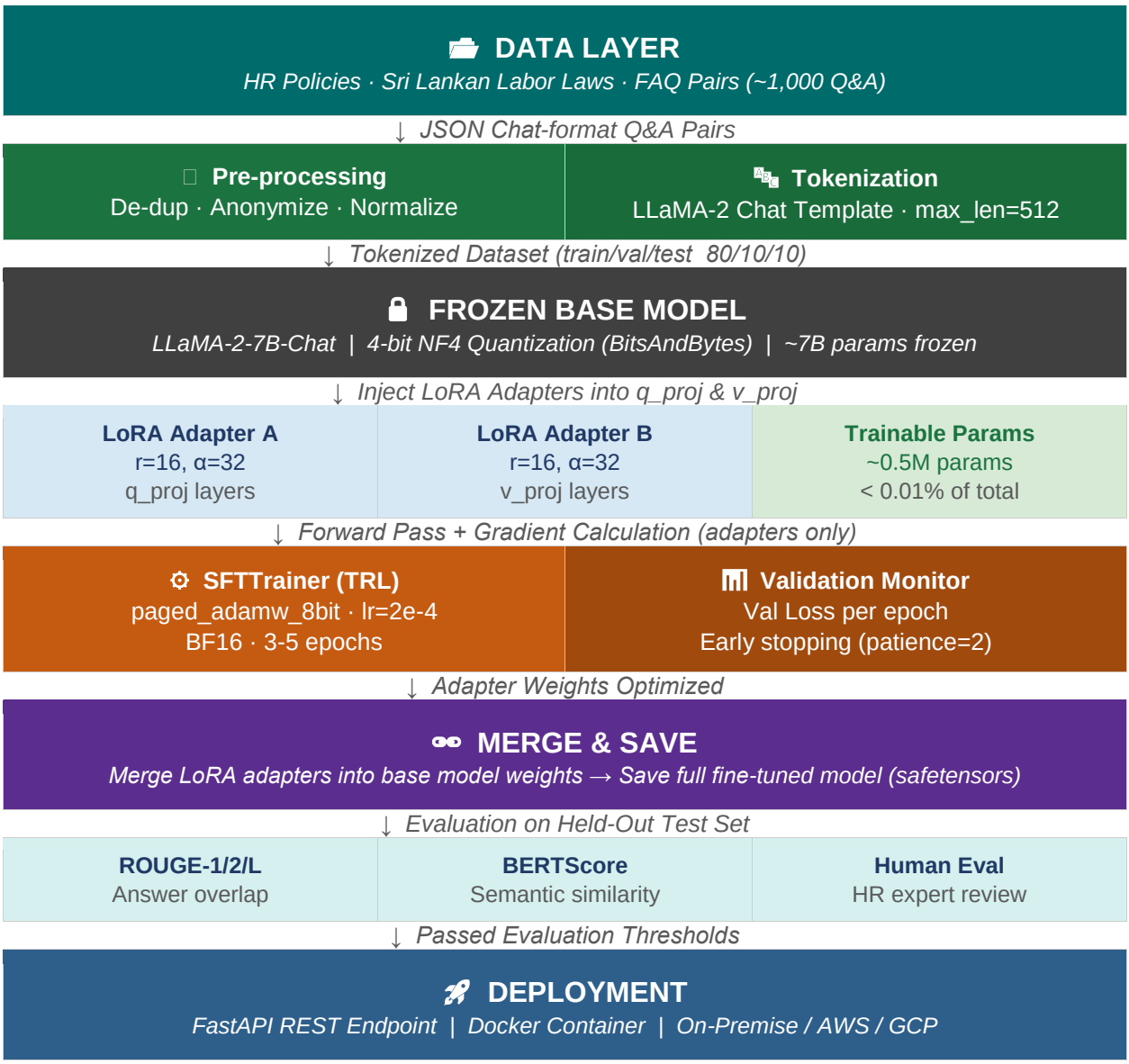
From an engineering and business standpoint, cost efficient (no per token API usage), fully private, scalable, technically robust.

This QLoRA based fine-tuning strategy ensures that LLaMA-2 is effectively adapted to the HR domain while maintaining infrastructure efficiency and enterprise compliance standards.

# 6 Architecture Diagram

The diagram below illustrates the complete end-to-end pipeline for QLoRA fine-tuning of LLaMA-2 for the HR Policy Assistant use case, from raw data ingestion through to deployment. Each layer represents a distinct stage in the pipeline, with data flowing top-to-bottom.

**QLoRA Fine-Tuning Architecture — LLaMA-2 HR Policy Assistant**



# 7. Training Pseudo-Code

https://github.com/Chamindu77/QLoRA-Fine-Tuning-LLaMA-2-7B-Chat-for-HR-Policy-Assistant

# 8. Evaluation Strategy

A robust, multi-dimensional evaluation strategy is essential to validate that the fine-tuned LLaMA-2 HR Policy Assistant meets both technical performance benchmarks and real-world HR accuracy requirements. The evaluation combines automated metrics with human expert review.

## 8.1 Evaluation Metrics

| Metric | Type | What It Measures |
|---|---|---|
| **ROUGE-1** | Automated | Unigram overlap between generated and reference answer |
| **ROUGE-2** | Automated | Bigram overlap captures phrase level accuracy |
| **ROUGE-L** | Automated | Longest common subsequence structural similarity |
| **BERTScore F1** | Automated | Semantic similarity via BERT embeddings |
| **Policy Accuracy** | Human | HR expert verifies factual correctness vs labor law |
| **Hallucination Rate** | Human | % of responses containing fabricated or incorrect facts |
| **Tone Compliance** | Human | Formal corporate tone maintained throughout response |
| **Perplexity (PPL)** | Automated | Lower PPL = more confident, fluent text generation |

## 8.2 Baseline Comparison

The fine-tuned model is evaluated against the unmodified LLaMA-2-7B-chat baseline (with a system prompt but no fine-tuning) to quantify the improvement attributable to domain adaptation.

# 9 Deployment Strategy

The fine-tuned HR Policy Assistant model is deployed as a production-grade REST API service, designed for enterprise environments. The deployment is containerized for portability and scalable from single-server on-premise deployments to cloud-hosted Kubernetes clusters.

| **Component** | **Technology** |
|---|---|
| Model Serving | FastAPI + Uvicorn |
| Containerization | Docker |
| Orchestration | Kubernetes / Docker Compose |
| Inference Engine | vLLM / llama.cpp |
| API Gateway | Nginx / AWS API Gateway |
| Monitoring | Prometheus + Grafana |
| Logging | ELK Stack / CloudWatch |
| Cloud Option | AWS / GCP / Azure |

# 10 Conclusion

The decision to use LLaMA-2 over GPT-3.5 Turbo is justified across all critical dimensions. LLaMA-2 provides full fine-tuning control and weight access, enabling true domain adaptation rather than superficial prompt engineering. It supports on-premise deployment for maximum data privacy compliance, incurs no per-token inference costs at production scale, and allows complete architectural customization including integration with retrieval systems, custom tokenization, and quantization. For an enterprise HR use case involving sensitive employee data and Sri Lanka-specific legal requirements, LLaMA-2 flexibility and deployability make it the clearly superior choice.

In conclusion, this assessment demonstrates that a thoughtfully designed QLoRA fine-tuning pipeline for LLaMA-2, grounded in high-quality Sri Lankan HR policy data, delivers a production-ready, accurate, private, and cost effective HR assistant solution. The combination of technical rigor, business alignment, and a clear deployment and maintenance strategy positions this solution as a robust foundation for enterprise AI adoption in the HR domain.